QUESTION BANK

Subject: Foundation of AI and Automation

Module 1 • Total Questions: 30 (7 Short | 13 Medium | 10 Long)

QUES. NO.	CO'S	ANSWER LENGTH	MARKS	STATEMENT OF THE QUESTION
1	1	Short	2	Define Artificial Intelligence (AI). Explain its scope and major applications.
2	1	Short	2	What is the Turing Test? Explain its significance in Artificial Intelligence.
3	1	Short	2	What is meant by Problem Reduction in Artificial Intelligence? Explain with a suitable example.
4	1	Short	2	Define State Space Search. Explain the different components of a state space.
5	1	Short	2	What are Intelligent Agents? Explain the basic structure of an intelligent agent.
6	1	Short	2	Define an agent and its environment. Explain their interaction with a suitable example.
7	1	Short	2	What is Heuristic Search? Why are heuristic functions used in AI?
8	1	Medium	5	Explain the historical development and evolution of Artificial Intelligence.
9	1	Medium	5	Explain the Chinese Room Argument. How does it challenge the idea of machine intelligence?
10	1	Medium	5	Differentiate between the Turing Test and the Chinese Room Argument.
11	1	Medium	5	Explain the concept of Problem-Solving Algorithm in AI with a suitable example.
12	1	Medium	5	Explain the major components of an Artificial Intelligence system.
13	1	Medium	5	What is a Search Process in AI? Explain its major steps.
14	1	Medium	5	Explain the concept of rational agent. What makes an agent rational?
15	1	Medium	5	Explain different types of environments in Artificial Intelligence.
16	1	Medium	5	What is a Problem-Solving Agent? Explain its working process.
17	1	Medium	5	What is Brute Force Search? Explain its working, advantages, and limitations.
18	1	Medium	5	Explain Depth First Search (DFS) as a search technique in Artificial Intelligence.
19	1	Medium	5	Explain Breadth First Search (BFS) with a suitable example.

QUES. NO.	CO'S	ANSWER LENGTH	MARKS	STATEMENT OF THE QUESTION
20	1	Medium	5	Differentiate between Breadth First Search and Depth First Search.
21	1	Long	10	Explain the relationship between Problem Reduction, State Space Search, and Problem Solving.
22	1	Long	10	Discuss the characteristics of an environment such as fully observable/partially observable, deterministic/stochastic, and static/dynamic.
23	1	Long	10	Explain the different steps involved in problem formulation by a problem-solving agent.
24	1	Long	10	Compare BFS and DFS based on completeness, optimality, time complexity, and space complexity.
25	1	Long	10	Explain the Hill Climbing Algorithm with a suitable example. Discuss its limitations.
26	1	Long	10	What are local maximum, plateau, and ridge in the Hill Climbing algorithm? Explain strategies to overcome them.
27	1	Long	10	Explain the Best First Search Algorithm. How does it differ from uninformed search?
28	1	Long	10	Explain the A* search algorithm. What are the roles of g(n), h(n), and f(n)?
29	1	Long	10	Explain the AO* algorithm. How is AO* different from A* search?
30	1	Long	10	What is Knowledge Representation in Artificial Intelligence? Explain its importance and major approaches.